

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : 3

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1. Problem Title : Hospital Appointment Management System

2. Problem Description : In many hospitals, appointments are still booked manually, which often leads to long waiting times and scheduling conflicts. Patients may struggle to find available time slots, and doctors may face overlapping appointments.

This system aims to make the process easier by allowing patients to book appointments based on doctor availability. Doctors can manage their schedules, and receptionists can help organize and update bookings.

All appointment details and medical records are stored digitally, reducing errors and improving efficiency. Overall, the system helps provide a smoother experience for both patients and hospital staff.

3. Identified Entities (Classes)

Class Name	Description
Patient	Represents a person seeking medical consultation
Doctor	Represents a medical professional who provides consultation
Receptionist	Manages appointment scheduling and coordination
Appointment	Represents a scheduled meeting between patient and doctor
Schedule	Maintains doctor availability and time slots
MedicalRecord	Stores patient visit history and treatment details
HospitalSystem	Core system managing all operations

4. Class Attributes and Behaviours

Patient Class:

Attributes	Methods(behaviours)
patientId	bookAppointment()
name	viewAppointments()
age	cancelAppointment()
contactNumber	updateDetails()

Doctor Class:

Attributes	Methods(behaviours)
doctorId	setAvailability()
name	confirmAppointment()
specialization	viewSchedule()
availability	updateAvailability()

Receptionist Class:

Attributes	Methods(behaviours)
receptionistId	scheduleAppointment()
name	manageAppointments()
contactNumber	updateSchedule()

AppointmentClass:

Attributes	Methods(behaviours)
appointmentId	createAppointment()
date	cancelAppointment()
time	rescheduleAppointment()
status	checkConflict()

Schedule Class:

Attributes	Methods(behaviours)
scheduleId	addTimeSlot()
availableSlots	removeTimeSlot()
doctorId	checkAvailability()

MedicalRecord Class:

Attributes	Methods(behaviours)
recordId	addRecord()
patientId	updateRecord()
diagnosis	viewRecord()
prescription	deleteRecord()

HospitalSystem Class:

Attributes	Methods(behaviours)
systemId	managePatients()
database	manageDoctors()
records	generateReports()
users	maintainRecords()